

RF GaN HEMT Compact Modelling with High-Temperature Effects for Extreme Environment Applications

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I. INTRODUCTION

Gallium nitride (GaN) high-electron-mobility transistors (HEMTs) are perceived as a frontrunner for communication and sensing electronics to maintain reliable operation in extreme environments (EE) such as wildfires, oil drilling, aerospace and space exploration [1]. Electronics are often required to endure temperatures ($T > 200^\circ\text{C}$) significantly above conventional operating ranges, hence requiring process optimization, and specialized design and modeling infrastructure. NRC has made significant progress in developing a GaN process for EE [2]. Meanwhile, existing GaN compact models are primarily optimized for commercial and industrial applications (up to 185°C), and do not adequately capture all the high-temperature (high-T) effects. Indeed, as the operating temperature approaches several hundred degrees, device behaviour deviates substantially from conventional models due to carrier mobility degradation, self-heating and numerous parasitics. These effects reduce the predictive accuracy in existing models and limit their suitability for EE designs. Using the industry-standard MVSG model as a starting point, this work presents the model modification, calibration and fitting procedure for a high-T GaN HEMT compact model validated up to 500°C , serving as a first step towards MVSG for EE development.

II. METHODOLOGY

High-T MVSG Model: The basic model framework for the channel current/charges and gate leakage follows the industry-standard MVSG GaN HEMT model formulation [3]. As the temperature increases, the carrier mobility, injection velocity, and series resistance degrade due to increased phonon scattering, thereby reducing the drain current. The threshold voltage shifts primarily due to the bandgap. Gate leakage also increases substantially with temperature due to strongly thermally activated recombination currents. Self-heating introduces an additional bias-dependent temperature rise affecting both channel and gate leakage, which primarily affects high drain biases where power dissipation is greatest. In this proposed model, all these key physical effects are identified, examined and analytically captured (Table 1).

High-T Parameter Extraction Flow Development: A systematic parameter extraction flow, critical for the practical adoption of any device compact model, is also developed, as sequenced in Table 1. The parameter extraction at t_{nom} (Step 0) is first performed following the standard MVSG parameter extraction flow [4]. Temperature coefficients are then extracted with the following (Steps 1-6) in Table 1.

III. RESULTS

The model is validated against measured data for NRC 1st generation EE HEMTs from 22°C to 500°C . Output characteristics (I_D - V_{DS}) and transfer characteristics (I_D - V_{GS}) demonstrate good agreement between model and measurement across all temperature conditions (Fig. 1), capturing the observed reduction in drain current. Gate-leakage characteristics are reproduced across multiple drain bias conditions. In particular, at elevated drain biases, the bias-dependent self-heating effect is captured through the thermal subcircuit, reproducing the measured spread in the (I_G - V_{GS}) plots across different drain bias conditions.

Table 1. Proposed High-T MVSG Parameter Extraction Flow

Steps	Target physical quantities and equations related with their temperature dependence	Required measurement	Extracted tempco
0	MVSG extraction at t_{nom} following standard MVSG parameter extraction flow [1]	N/A	N/A
1	Series resistance $R_{ctDUT} = R_{cnom} \cdot (1 + t_{c,Rc} \cdot (t_{DUT} - t_{nom}))$ (1) $R_{shDUT} = R_{shnom} \cdot (1 + t_{c,Rsh} \cdot (t_{DUT} - t_{nom}))$ (2)	TLM measurement	$t_{c,Rc}$, $t_{c,Rsh}$
2	Threshold voltage $V_T(t_{DUT}) = V_{Tnom} + t_{c,VT} \cdot (t_{DUT} - t_{nom})$ (3)	I_D vs. V_{GS} at low V_{DS}	$t_{c,VT}$
3	Mobility $\mu = \frac{\mu_{nom}}{(1 + t_{c,\mu} \cdot (t_{DUT} - t_{nom}))}$ (4)	I_D vs. V_{GS} at low V_{DS}	$t_{c,\mu}$
4	Saturation velocity $V_{so} = \frac{V_{so, nom} (1 + t_{c,vso} \cdot t_{nom})}{(1 + t_{c,vso} \cdot t_{DUT})}$ (5)	I_D vs. V_{DS}	$t_{c,vso}$
5	Gate leakage forward bias inverse non-ideality factor: $p_{gs}(t_{DUT}) = p_{gsnom} (1 + t_{c,pgs} \cdot (t_{DUT} - t_{nom}))$ (6)	I_D vs. high V_{GS}	$t_{c,pgs}$, $t_{c,pgd}$
6	Gate leakage reverse-recombination inverse non-ideality factor: $p_{gs,rec}(t_{DUT}) = p_{gs,rec, nom} (1 + t_{c,pgs,rec} \cdot (t_{DUT} - t_{nom}))$ (7)	I_D vs. low V_{GS}	$t_{c,pgs,rec}$, $t_{c,pgs,rec,d}$

Note: The device temperature (t_{DUT}) is calculated as $t_{DUT} = t_{amb} + dt$. Here t_{amb} is the ambient temperature and dt is the temperature change due to device self-heating computed from a first-order thermal R_{th} - C_{th} network. t_{DUT} is used to compute the key parameters affecting channel current (Eq. (1) – (5)) and gate-leakage (Eq. (6) – (7)). t_{nom} is the nominal temperature, usually set to be room temperature (RT).

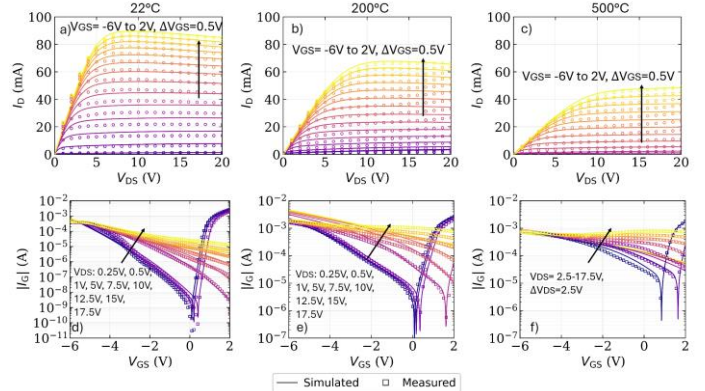


Fig. 1. Measurement (symbols) vs high-T MVSG model (lines) results for NRC 1st generation EE GaN HEMTs, I_D - V_{DS} at (a) 22°C , (b) 200°C and (c) 500°C , and I_G - V_{GS} at (d) 22°C , (e) 200°C and (f) 500°C .

IV. CONCLUSION

This work models the high-T behaviours of GaN HEMTs, extending predictive accuracy of the existing MVSG model up to 500°C . A corresponding systematic parameter extraction flow is also developed. Good agreement with experimental measurements validates the proposed model and extraction flow. This work provides a practical framework for circuit design targeting extreme environment GaN applications.

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References: [1] C. Wang *et al.*, *IEEE Trans. Microw. Theory Techn.*, 2014. [2] H. Xue *et al.*, in *CCECE*, 2024. [3] L. Wei *et al.*, in *BCICTS*, 2024. [4] R. Fang *et al.*, in *BCICTS*, 2025